XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Week 11
Lab Practice

Write a C code to find the first derivative (dy/dx) and the second derivative (d^2y/dx^2) at each point in the following table using the forward-difference method.

x	0	0.25	0.5	0.75	1.0	1.25	1.5
y	−1	1	2.5	4	2	1.5	1

Then, write the C code to implement the backward-difference rule and the central-difference rule to find the first derivative and the second derivative of the points in the table above.

// Beginning of the C Source code

Function main()

```
1   // forward difference rule
2   #include <stdio.h>
3
4   int main()
5   {
6       float x[7] = {0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5};
7       float y[7] = {-1, 1, 2.5, 4, 2, 1.5, 1};
8       float dydx[7] = {0.0}; // store the first derivative values
9       float d2ydx2[7] = {0.0}; // store the second derivative values
10      float h = x[1] - x[0];
```

Line 6 to 7: Declare and initialize x and y as given in the question.

Line 8: Declare an array dydx to store the first derivative results.

Line 9: Declare an array d2ydx2 to store the second derivative results.

Line 10: Declare a variable h which is the distance between two consecutive x values.

Forward Difference Code

```
12      for(int i=0; i<7; i++)
13      {
14          if(i!=6)
15              dydx[i] = (y[i+1] - y[i])/h;
16          if(i!=6 && i!=5)
17              d2ydx2[i] = (y[i+2] - 2.0*y[i+1] + y[i])/(h*h);
18          printf("\nx=%f, y=%f, y'=%f, y''=%f", x[i], y[i], dydx[i], d2ydx2[i]);
19      }
20      printf("\n");
21      return 0;
22  }
```

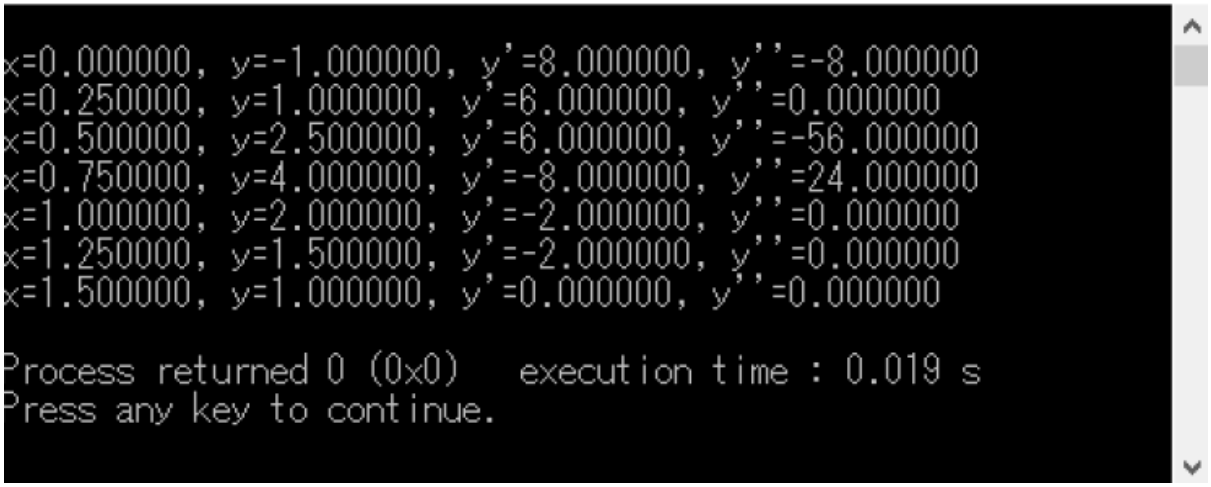
Line 15: Implement the forward difference rule equation for the first derivative in C code:

$$y_i' \approx \frac{y_{i+1} - y_i}{h}.$$

Line 17: Implement the forward difference rule equation for the second derivative in C code:

$$y_i'' \approx \frac{y_{i+2} - 2y_{i+1} + y_i}{h^2}.$$

Output


x=0.000000, y=-1.000000, y'=8.000000, y''=-8.000000
x=0.250000, y=1.000000, y'=6.000000, y''=0.000000
x=0.500000, y=2.500000, y'=6.000000, y''=-56.000000
x=0.750000, y=4.000000, y'=-8.000000, y''=24.000000
x=1.000000, y=2.000000, y'=-2.000000, y''=0.000000
x=1.250000, y=1.500000, y'=-2.000000, y''=0.000000
x=1.500000, y=1.000000, y'=0.000000, y''=0.000000
Process returned 0 (0x0)   execution time : 0.019 s
Press any key to continue.